C-STGB: Conformal Spatio-Temporal GraphBoost End-to-End Surveillance Platform

STAGE 1: STREAMING INGEST
DuckDB Invariants & Hawkes

Continuous Ingest & Subgraphs

$$G_t = (V_t, E_t), \quad N_K(u)\ (K \le 15)$$

DuckDB / Arrow zero-copy streaming

Mass Flow Conservation

$$\Phi_{\text{flow}} = \log(1 + A_{\text{out}}/[A_{\text{in}} + \varepsilon]) \approx 1.0$$

Layering dissipation & pass-through detector

Hawkes Intensity & Causal Taint

$$\lambda_u(t) = \mu_u + \sum \alpha \cdot e^{-\beta \Delta t}$$

Forward & backward taint (< 0.08 ms)

Out: $\mathbf{z}_{\text{inv}}(u,t) \in \mathbb{R}^{12},\ G_t^{(K)}$

0.45 ms SLA | DuckDB Ingest

STAGE 2: GNN & FILTER
Tri-Band & Latent GraphSMOTE

Tri-Band Temporal Attention

$$\Phi_{\text{time}}(\Delta t),\ w(\Delta t) = \sum \pi_b e^{-\lambda_b \Delta t}$$

Microsecond bursts to 90-day dormancy

Learnable Edge-Trust Gating

$$\hat{g}_{ij} = \delta_{\text{floor}} + (1 - \delta_{\text{floor}}) g_{ij}$$

Prunes 65.9% of adversarial chaff links

Typology Latent GraphSMOTE

$$h_{\text{syn}} = (1 - \rho) h_u + \rho h_v,\ u, v \in C_k$$

Latent interpolation & hard-negative pairs

Out: $\mathbf{h}_u^{(L)} \in \mathbb{R}^d$ (Latent Embedding)

Continuous Temporal Dynamics

STAGE 3: FUSION & RISK
Evidence-Adaptive Bayes Head

Evidence-Adaptive Fusion Gate

$$\alpha_u = \sigma(\mathbf{w}_\alpha^\top [\mathbf{h}_u \parallel \mathbf{z}_{\text{inv}}] + b_\alpha)$$

Cold-start: deg(u) <= 2 shifts to z_inv

Unified Representation Space

$$\mathbf{h}_u^* = \alpha_u \mathbf{h}_u + (1 - \alpha_u)[\mathbf{z}_{\text{inv}} \parallel \mathbf{x}_u]$$

Secures 89.6% F1 on isolated entities

Cost-Sensitive Bayes Head

$$\tau^* = \arg\min_\tau (15 \cdot \text{FN} + 1 \cdot \text{FP})$$

Asymmetric loss policy (C_FN / C_FP = 15)

Out: $\hat{p}_u \in [0,1],\ \mathbf{h}_u^* \in \mathbb{R}^{d^*}$

Cold-Start Guarded + Bayes Risk

STAGE 4: CONFORMAL RISK CONTROL
Pure CRC & Streaming Triage

Class-Conditional CRC

$$\hat{q}^{(y)} = \text{Quantile}_{1-\alpha}(D_{\text{cal}}^{(y)}),\ 1 - \alpha \ge 99.0\%$$

Finite-sample marginal & class risk bounds

3-Tier Deterministic Triage

$$\Gamma(u) \in \{\{1\},\ \{0,1\},\ \{0\}\}$$

Routes >99.4% to straight-through tiers

Adaptive Conformal Inference

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$$

Online Martingale TV drift adaptation

Out: $\Gamma(u) \subseteq \{0,1\},\ \hat{q}^{(0)}, \hat{q}^{(1)}$

Finite-Sample Coverage (>=99.0%)